



CAD Craft

Competition Rulebook

1.

Event Type:

Individual Event

Team Size:

1 Member (Software: FreeCAD)

Description:

CAD Craft is an individual CAD modelling competition designed to test participants' creativity, mechanical drafting, assembly knowledge, and dimensional precision. The competition is conducted using FreeCAD and consists of two intensive rounds: Creative CAD Modelling and Reverse Engineering.

Schedule & Rounds:

- Round 1: Creative CAD Modelling – 1 Hour 30 Minutes
- Break: 15 Minutes
- Round 2: Reverse Engineering & Precision – 1 Hour 15 Minutes

Round 1 – Creative CAD Modelling:

- Participants will be given a design challenge and must create an original CAD model during the allotted time.
- All required components must be properly modelled, assembled, and appropriately constrained.
- Participants must work independently; pre-existing or downloaded models are strictly prohibited.

Round 1 Evaluation (50 Marks):

- Creativity & Originality: 15 Marks | CAD Modelling Quality: 15 Marks
- Assembly & Constraints: 10 Marks | Accuracy & Attention to Detail: 10 Marks

Contacts:

Event Coordinators & Contact:

Coordinator 1: Ankita Thakur (+91 9816229682)

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Round 2 – Reverse Engineering & Precision:

- Participants will be provided with a 3D-printed physical model and will recreate it using FreeCAD.
- Participants may inspect and measure the given physical model.
- Accuracy of dimensions, shape, geometry, and assembly will be strictly evaluated.
- No external references, online models, or outside assistance will be allowed.
- Participants must complete the recreation within the allotted 1 Hour 15 Minutes.

Round 2 Evaluation (50 Marks):

- Dimensional Accuracy: 25 Marks
- Shape & Geometry Accuracy: 10 Marks
- Assembly Accuracy: 10 Marks
- Constraints & Model Organization: 5 Marks

General Rules & Lab Conduct:

- All participants will work on the systems available in the lab running FreeCAD.
- FreeCAD will be the designated software used for both rounds.
- This is an individual competition. Collaboration or communication between participants is not permitted.
- Use of unauthorized external resources, pre-existing models, or unfair means will result in disqualification.
- Participants must follow all instructions given by the judges.
- The decision of the judges will be final and binding.

Final Score:

- Round 1 (50 Marks) + Round 2 (50 Marks) = Total: 100 Marks